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Evaluation of Pulse Compression Techniques for X-band Radar Systems

V. Ankarao,[1] S. Srivatsa[2] and G.A. Shanmugha Sundaram[3]
[1,2,3]Center for Computational Engineering and Networking, Amrita School of Engineering, Coimbatore,
Amrita Vishwa Vidyapeetham, Amrita University, India
[3]Department of Electronics and Communications Engineering, Amrita School of Engineering, Coimbatore,
Amrita Vishwa Vidyapeetham, Amrita University, India
Email: [3]ga_ssundaram@cb.amrita.edu

***Abstract*—Detection of moving targets using a moving target indication (MTI) type radar system has two important tasks viz., determination of the precise range and the velocity of the target. Precise range calculations require the radar system to have a high range resolution ability. An important trade-off factor in radar systems is between range resolution, that requires shorter pulse, and transmitted pulse energy, that necessitates longer pulses. Pulse compression is a technique that helps overcome this trade-off by modulating the pulses to be transmitted. Various frequency and phase modulation techniques can be used to achieve compression of the pulses. In the discussions here, the linear frequency modulation (LFM) and non-linear frequency modulation (NLFM) methods are compared with phase modulation techniques such as Barker codes and polyphase codes. Two new approaches for pulse compression that employ quadrature phase-shift keying (QPSK) and quadrature amplitude modulation (QAM) are also presented. These pulse compression techniques are compared for relative performance using parameters such as the peak-sidelobe ratio (PSLR), pulse compression ratio and Doppler tolerance. Results show that polyphase codes have high Doppler tolerance. QAM and QPSK, like polyphase codes, are not limited by number of bits that can be used within a pulse, and thus can be used to achieve good compression ratios.**

***Index Terms*—Pulse compression, Range resolution, Barker codes, Polyphase codes, PSLR, Doppler tolerance.**

I. Introduction

Radars are used in various applications, from air and sea traffic control to threat target detection. They are also used in atmospheric monitoring, precise distance and speed measurements of land vehicles, industrial and bio-medical applications [1]. Although radars are versatile and are used in various fields with different functioning parameters, the basic working principle is the same. A known signal is transmitted and the echoes reflected off the targets are analyzed. The most common and an important application of radar is detection of flying objects, such as passenger flights, fighter jets, missiles etc. Moving target indication (MTI) radars are used for the detection of these types of targets [2]. MTI radars operated with the principle that a moving target induces a Doppler shift in the reflected echoes, while the stationary objects just reflect the signals without introduction of any Doppler shift. MTI filters make use of two/three-pulse cancellers to remove the echoes from stationary surfaces, which are termed as clutter.

When multiple targets are nearby to each other, it is required that the range resolution be high for them to be detected as separate targets. A radar system with low range resolution might not show up different targets which are close-by. High range resolution demands that the pulse width must be shorter, but a shorter pulse will have low pulse energy, thus reducing the signal-to-noise ratio (SNR) of the echoes of the transmitted pulses [3]. Various techniques for improving the SNR of the received signals have been proposed. Coherent detection is one such important technique, where multiple pulses are coherently summed. Since echoes are coherent, the summed signal will have high magnitude, but noise in most cases is non-coherent and ideally adds to zero when coherently summed. So, SNR of the received signal is increased to a great extent. But, presence of more noise necessitates having high pulse energy. A peak power in pulses requires higher transmission power, which is often desirable. So, there is a need for techniques that can provide high range resolution without reducing the width of the pulses. Pulse compression is one technique that can solve this issue.

In the receiver part of a radar system, received signals are always passed through a matched filter, so that the detectability of the signals in the presence of noise is increased. But, passing a rectangular pulse through a matched filter produces a triangular output. However, signals having a large time-bandwidth product produce a narrow impulse with a large peak value. Thus, it‘s possible to obtain pulse compression by modulating the transmitted pulses. Modulation schemes, such as linear frequency modulation (LFM), non-linear frequency modulation (NLFM) and phase modulation techniques can be found in literature [4–6] and also have been implemented in practical radar systems. Phase modulation techniques like binary phase modulation, Barker codes, and polyphase codes are also used in radars.

In this paper, these compression techniques are compared for their performance with following parameters: peak sidelobe ratio (PSLR), compression ratio and Doppler tolerance. Pulse compression can also be achieved with quadrature amplitude modulation (QAM) and quadrature phase shift keying (QPSK). Effectiveness of using these techniques are compared with previously mentioned existing techniques.

The paper is organized as: Section II gives a description of LFM and NLFM techniques. Section III briefly explains phase modulation based pulse compression techniques. Section IV

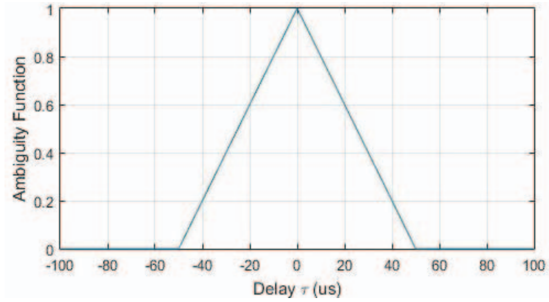

Fig. 1. Auto-correlation of a rectangular pulse.

discusses the results and comparison of the modulation techniques described in Sections II and III. Conclusion and future possibilities are given in Section V.

II. Frequency Modulation Based Pulse Compression

Modulation of a signal can increase the transmission bandwidth and when matched filtered at the receiver, a narrow peak is obtained. So, frequency and phase modulation schemes can be used to obtain compression of radar pulses. In an ordinary frequency modulation, auto-correlation of the signal gives high energy sidelobes. So, a frequency modulated signal in which, frequency varies with time may be used to obtain highly suppressed sidelobes, so that a good pulse compression may be achieved [7]. In regard of frequency-time relationship, frequency modulation can be either linear or non-linear and are briefly discussed here.

Fig. 1 shows the auto-correlation output of a rectangular pulse. It is clear from the plot that after matched filtering of the received pulses, the output occupies complete time period and so, two or more targets lying close enough to appear within a coherent processing interval (CPI) will not be shown up in the receiver. But, if the output of the matched filter has a narrow peak, then multiple targets can be detected that are very close enough, depending on the time bandwidth of the narrow peak.

A. Linear Frequency Modulation (LFM)

The utility of using frequency modulation techniques is that they can be readily generated with various techniques [4]. A rectangular pulse modulated with a sine wave may be represented as:

$$x\,(t) = rect(t/T_r)A\cos\left(2\pi f_0 t + \varphi\right)$$

where, *rect* is a rectangular function, which is defined by:

$$rect\left(\frac{t}{T_r}\right) = \begin{cases} 1, & |t| \leq T_r/2 \\ 0, & |t| \geq T_r/2. \end{cases}$$

Here, 'A' is the amplitude, f_0 is the frequency, and φ is the initial phase of the signal. Here, f_0 is constant. Suppose, this term is made variable with respect to time, then a chirp signal is obtained. A chirp is a signal which sweeps a range of frequencies over a given time period. Based on the relationship between time and frequency, the modulation can be linear or non-linear frequency modulation [8]. A chirp signal can be described as:

$$x\,(t) = rect(t/T_r)A\cos\left(2\pi f_i(t)t + \varphi\right).$$

Here, the properties of $f_i(t)$ decides the modulation type and performance. A linear relationship between frequency and time can be obtained by $f_i\,(t) = \alpha t + f_0$, the value of α being $\alpha = (f_1 - f_0)/2T$, where T is the total sweep duration. f_0 is the initial frequency of the sweep and f_1 is the frequency at the end of the sweep.

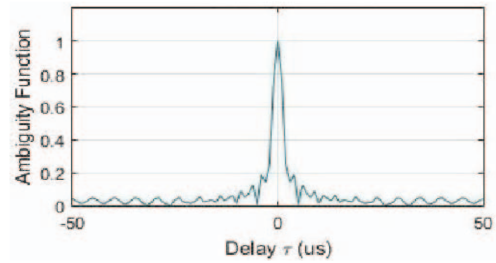

Fig. 2. Auto-correlation of LFM.

Fig. 2 shows the auto-correlation output when LFM signal is used instead of a rectangular pulse. Clearly, it can be observed that a central peak is obtained with LFM, unlike a rectangular pulse. Thus, the effect due to modulation of transmitted pulses in order to achieve pulse compression can be verified.

B. Non-Linear Frequency Modulation (NLFM)

The linearity plot of LFM shows the variation of frequency with time is linear. A various non-linear terms may be introduced in the equation of $f_i\,(t)$ to make the time-frequency relation non-linear. Some of the simplest ideas to make it non-linear are by introducing a cosine function, exponential term, quadratic or higher order terms etc. Introduction of a quadratic term with an arbitrary value β makes the equation of $f_i\,(t)$ as [6]:

$$f_i\,(t) = \beta t^2 + \alpha t + f_0.$$

It becomes clear from the above equation that a term with t^2, which is a non-linear term makes the output function non-linear. So, ideally a non-linear frequency modulated signal is obtainable with the introduction of quadratic term.

The LFM and NLFM generate sidelobes and can be reduced with the help of amplitude weighting techniques. Also, these techniques are more susceptible to Doppler shift. Introduction of Doppler shift in the return echoes reduces the main lobe to a greater extent and increases peak- sidelobe ratio (PSLR) value.

III. Phase Modulation Based Pulse Compression

Like modulation using sine signals, radar pulses can also be modulated with various phase changes to increase bandwidth of the signal. Simplest way is to use binary phase shift with a code sequence. But, random code sequence is not effective. So, Barker codes were introduced which give equal

TABLE I
LIST OF EXISTING BARKER CODES WITH PSLR VALUES.

Length of code	Elements in code	PSLR (dB)
2	10 or 11	−6
3	110	−9.5
4	1101 or 1110	−12
5	11101	−14
7	1110010	−16.9
11	11100010010	−20.8
13	1111100110101	−22.3

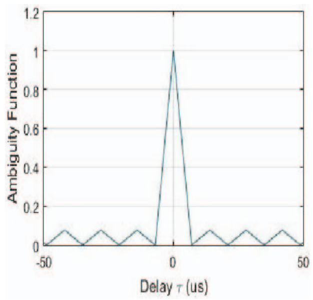

Fig. 3. Auto-correlation of 13-bit Barker code.

amplitude sidelobes [5]. Other than Barker codes, various polyphase signals have also been used. Some of the important ones are the P1, P2, P3, P4 and Frank codes [9]. In this paper, we discuss two new approaches for phase coded pulse compression. They are based on QPSK and QAM.

A. Barker Codes

Barker codes is a binary phase coded pulse compression technique. These codes are also called as perfect code sequences, as they give equal amplitude time sidelobes. The longest Barker code obtained is of 13-bits. Hunt for longer Barker codes have not provided any fruitful results. Advantage of these codes is that there is no need for amplitude weighting to suppress sidelobes.

Table I shows the list of existing Barker codes. It also gives the PSLR value obtainable with these binary codes [10], [11]. These PSLR values are obtained when there is no Doppler shift or delay introduced in the auto-correlation process.

The auto-correlation output for a Barker code of length 13 bits is shown in Fig. 3. It is observed from the plot that all the time sidelobes have equal amplitude. Although Barker codes provide equal time sidelobes, their biggest drawback is the amount of pulse compression achievable. The compression achievable by a phase code is directly related to the number of sub-pulses. Since the maximum number of bits available in Barker codes in 13, the maximum pulse compression achievable is 13 dB.

B. Polyphase Codes

Polyphase codes use harmonically related phases which are based on certain fundamental phase increment. Some of the important polyphase codes are Frank code, P1, P2, P3 and P4 codes. Frank, P1 and P2 codes are derivable from step approximations, while P3 and P4 are derivable from linear frequency modulation waveforms. Table II lists the equations applied to generate these polyphase codes. The codes have L^2 elements and the phase of the mth code element in the nth row of the code group is computed as given in the table.

TABLE II
GENERATION OF POLYPHASE CODES.

Polyphase code	Phase sequence for N bits
Frank	$\phi_{m,n} = \frac{2\pi}{L}(m-1)(n-1)$
P1	$\phi_{m,n} = \frac{-\pi}{L}\left[L-(2n-1)\right]\left[(n-1)L+(m-1)\right]$
P2	$\phi_{m,n} = \frac{\pi}{2L}\left[L-2m+1\right]\left[L-2n+1\right]$
P3	$\phi_{m,n} = \frac{\pi}{L}(m-1)^2$
P4	$\phi_{m,n} = \frac{\pi}{L}(m-1)(m-L-1)$

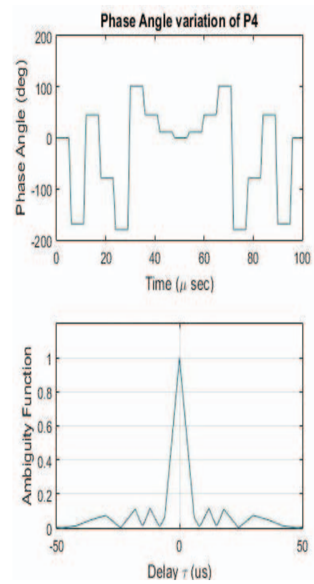

Fig. 4. P4 code and its auto-correlation.

These codes can have a length ‘L’ and so can produce much higher compression ratios compared to Barker codes. But, although it is required to have high bandwidth for obtaining more compression, we cannot arbitrarily choose high value of ‘L’ since the available bandwidth will be normally low for transmission of a signal. Also, electronic components will not be linear over full frequency spectrum to operate satisfactorily. Thus, the choice of length of polyphase codes is an important factor.

The phase angle variation in the P4 code and its auto-correlation function are plotted in Fig. 4. The length of P4 in this plot is 16 bits. It can be observed from auto-correlation output that the time sidelobes are already suppressed to a great extent, but are not of equal amplitude as like in Barker codes.

C. Quadrature Phase Shift Keying (QPSK)

QPSK is a form of m-ary PSK (phase shift keying) scheme, where $M = 4$. It has four points on the constellation diagram

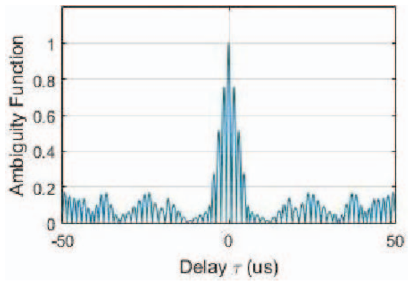

Fig. 5. Auto-correlation of QPSK signal.

which are equi-distant around a circle. They encode two bits in each symbol. QPSK signal can be mathematically described as:

$$s_n(t) = \sqrt{\frac{2E_s}{T_s}} \cos\left(2\pi f_c t + (2n-1)\frac{\pi}{4}\right),\ 1 \leq n \leq 4.$$

This gives out four phases as $\pi/4$, $3\pi/4$, $5\pi/4$, $7\pi/4$, and a two-dimensional signal space that consists of unit basis functions:

$$\phi_1(t) = \sqrt{\frac{2}{T_s}}\cos(2\pi f_c t)$$

$$\phi_1(t) = \sqrt{\frac{2}{T_s}}\sin(2\pi f_c t).$$

The first basis function of the signal represents an in-phase component and the second basis function is the quadrature-phase component. Since QPSK uses two bits per symbol representation, it provides double the data rate of BPSK and so is twice as efficient as binary phase systems. Similar to polyphase codes, modulation with QPSK can have any value of length of bits and is a choice of bandwidth required. Choice of the bit pattern is a crucial part in using QPSK for pulse compression.

A plot of the auto-correlation function for a 16-bit QPSK signal is shown in Fig. 5. Low sidelobes levels are visible in the figure. A different bit pattern produces different levels of sidelobes. A pattern that contains least power in the sidelobes is always preferable and there is need to search for such a bit pattern.

D. Quadrature Amplitude Modulation (QAM)

Being a mixture of a digital technique and an analog technique, QAM modulates two digital bit-streams or two analog message signals [12]. The modulated wave is a combination of PSK and ASK. Two carriers of same frequency are out of phase by π radians and so are called as quadrature carrier signals. These sinusoids are amplitude modulated with two modulating signals, then summed to obtain the final QAM signal [12]. The transmitted signal takes the form–

$$s(t) = A_I(t)\cos(2\pi f_0 t) - A_Q(t)\sin(2\pi f_0 t)$$

where, $A_I(t)$ and $A_Q(t)$ are modulating signals. The phase of the resulting periodic signal can be adjusted by changing amplitude of A_I and A_Q.

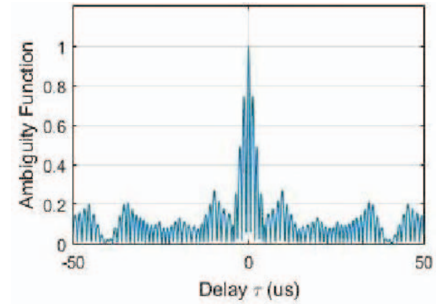

Fig. 6. Auto-correlation of 16-QAM signal.

Auto-correlation of QAM can be compared with other pulse compression technique from Fig. 6. A much narrower main lobe is observed in this plot. Similar to QPSK, these signals are not limited by the number of bits that can be used, but the choice of bit pattern plays an important role in the quality of pulse compression achieved.

IV. Results and Discussion

Important inferences about a pulse compression signal can be obtained with the help of auto-correlation function. Auto-correlation function is similar to matched filtering of a signal. In the previous section, auto-correlation outputs of various pulse compression techniques were presented. In this section, the effect of Doppler shift on the auto-correlation outputs will be discussed.

In the case of MTI radars, the radar receiver has to auto-correlate a signal with its Doppler shifted pulse sequence. Since, there is a phase lead/lag due to Doppler effect, the output of a matched filter will not be same as that of auto-correlation output. The main lobe of the auto-correlation function will have lower peak and energy gets more concentrated in the sidelobes. The resistance of a pulse compression technique to the Doppler effect is known as Doppler tolerance. Considering a radar system with parameters as: pulse repetition frequency (PRF) $= 10$ kHz; so, pulse repetition interval (PRI or T_p) $= 100$ µs. Assume a duty cycle of -20 dB, which gives pulse width $(\tau) = 1$ µs. The maximum unambiguous range is $R_{\max} = \frac{cT_p}{2}$, which results in an $R_{\max} = 15$ km. The range resolution obtainable without any pulse compression is $\Delta R = \frac{c\tau}{2} = 150$ m. Let the operating frequency of the radar be 10 GHz. So, a target with a velocity of 2 machs (660 m/s) produces a maximum Doppler shift of 22 kHz.

The variations in auto-correlation output for Barker 13-bit, P4, QPSK and QAM schemes are shown in Figs. 7–10 respectively, when they are subjected to a Doppler shift of 4 kHz It can be observed from the figures that LFM and NLFM are very much prone to Doppler effect. For low Doppler shifts, all the phase modulation schemes are less affected.

From the plots of auto-correlation when subjected to 4 kHz Doppler shift, we can infer the changes occurring over the main lobe. The peak of the main lobe relative to the sidelobes is reducing with the Doppler shift. It also affects the amount of pulse compression achieved, since introduction of Doppler

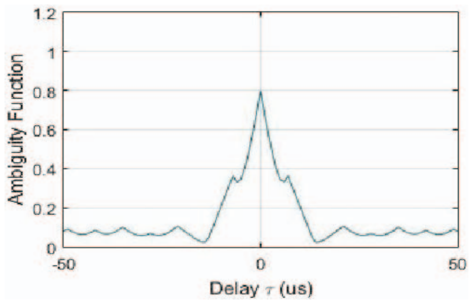

Fig. 7. Auto-correlation of Barker 13-bit with 4 kHz Doppler shift.

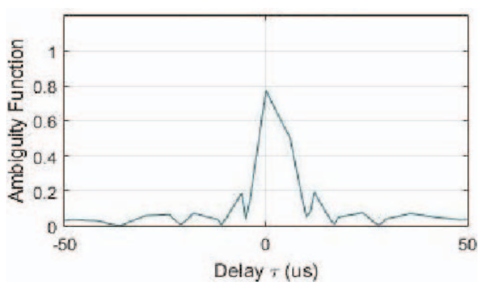

Fig. 8. Auto-correlation of P4 with 4 kHz Doppler shift.

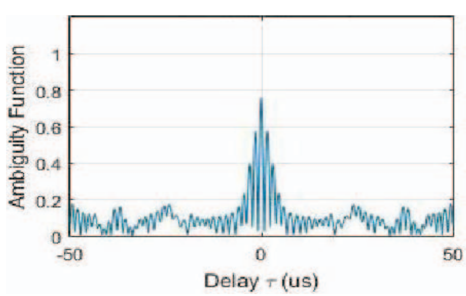

Fig. 9. Auto-correlation of QPSK with 4 kHz Doppler shift.

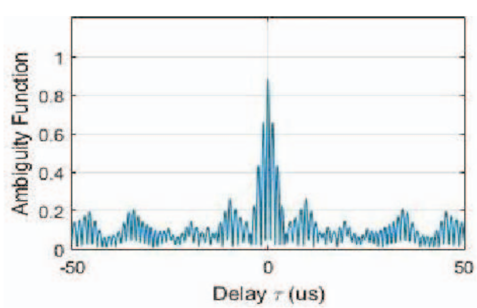

Fig. 10. Auto-correlation of QAM with 4 kHz Doppler shift.

shift or delay increases the time-width of the main lobe. Figs. 11 and 12 are plots of the variation in PSLR with Doppler shift. The polyphase codes P1 and P4 perform the best against Doppler shift, as compared to Barker 11 and 13 bits. But, QPSK and QAM are worse affected by the effect of Doppler. The choice of bit pattern for QPSK and QAM that are resistant to Doppler effect are required to be identified.

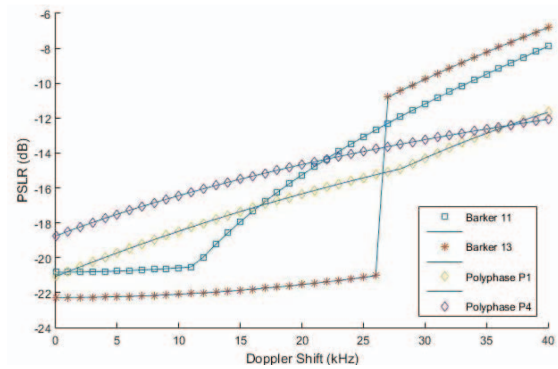

Fig. 11. PSLR vs. Doppler shift for Barker and polyphase codes.

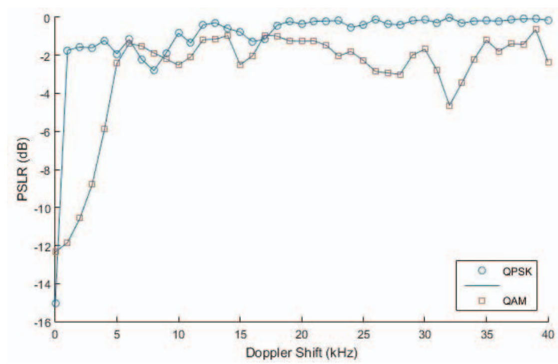

Fig. 12. PSLR vs. Doppler shift for QPSK and QAM.

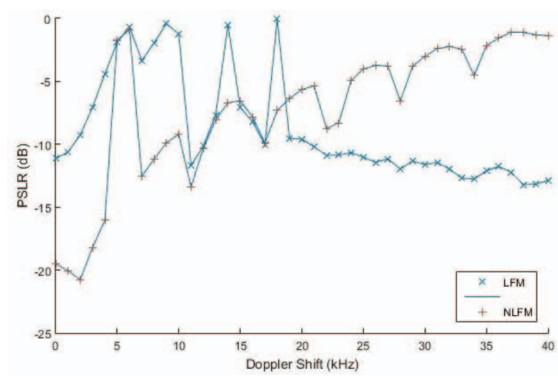

Fig. 13. PSLR vs. Doppler shift for LFM and NLFM.

LFM and NLFM are also very much affected by the Doppler shift introduced by moving targets. Variations in PSLR for LFM and NLFM are depicted in Fig. 13. There are large variations in the PSLR values of LFM and NLFM because with greater Doppler shifts there is a corresponding reduced presence of the main lobe, while two sidelobes with high peak values appear alongside.

V. Conclusions

The need for pulse compression in MTI radars have been discussed with description of various pulse compression techniques. Comparison of frequency modulation based pulse compression with phase modulation based pulse compression is also presented in terms of achievable PSLR value, pulse compression levels and effect of Doppler shift on the performance of these compression techniques. Two new pulse compression techniques are presented which are based on QPSK and QAM signal modulation techniques. Analysis of these techniques in terms of achievable compression ratios shows that all polyphase codes, QPSK and QAM are not limited by the number of sub-pulses that can be used, but the length of bits must be chosen based on the available bandwidth. In terms of Doppler effect, polyphase codes appear to be the most stable.

As a future work, precise code patterns for QPSK and QAM that are competitive with polyphase codes shall be identified. Recent modulation techniques such as the pulse amplitude modulation with 4 amplitude levels (PAM4) shall also be considered as a radar signaling scheme to achieve further enhancements in pulse compression while occupying reduced bandwidths, a definite advantage when it comes to digital processing of the radar signals.

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